

Deploy on GitHub Actions

Ticket change events. One poll. Then exit.

New trigger. Same exits. Same harness. The graph does not change.

Copy-me workflow: `labs/lab1_enhancer/workflows/enhance-on-issue.yml`

Notes: `labs/lab1_enhancer/GITHUB-ACTIONS.md`

Not Saturday. Copy onto **your CRM fork**. Never the instructor repo.

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What you will deploy

A workflow that GitHub already knows how to start.

```
on:
  issues:
    types: [opened, edited, labeled]
  issue_comment:
    types: [created]
  workflow_dispatch:
```

That is "the ticket changed": a new issue, an edited body, a label, or a human comment. LGTM arrives as `issue_comment`.

Why Actions instead of polling

`task poll-forever` dies when the laptop sleeps.

GitHub already emitted the event. Hosting a receiver is optional.

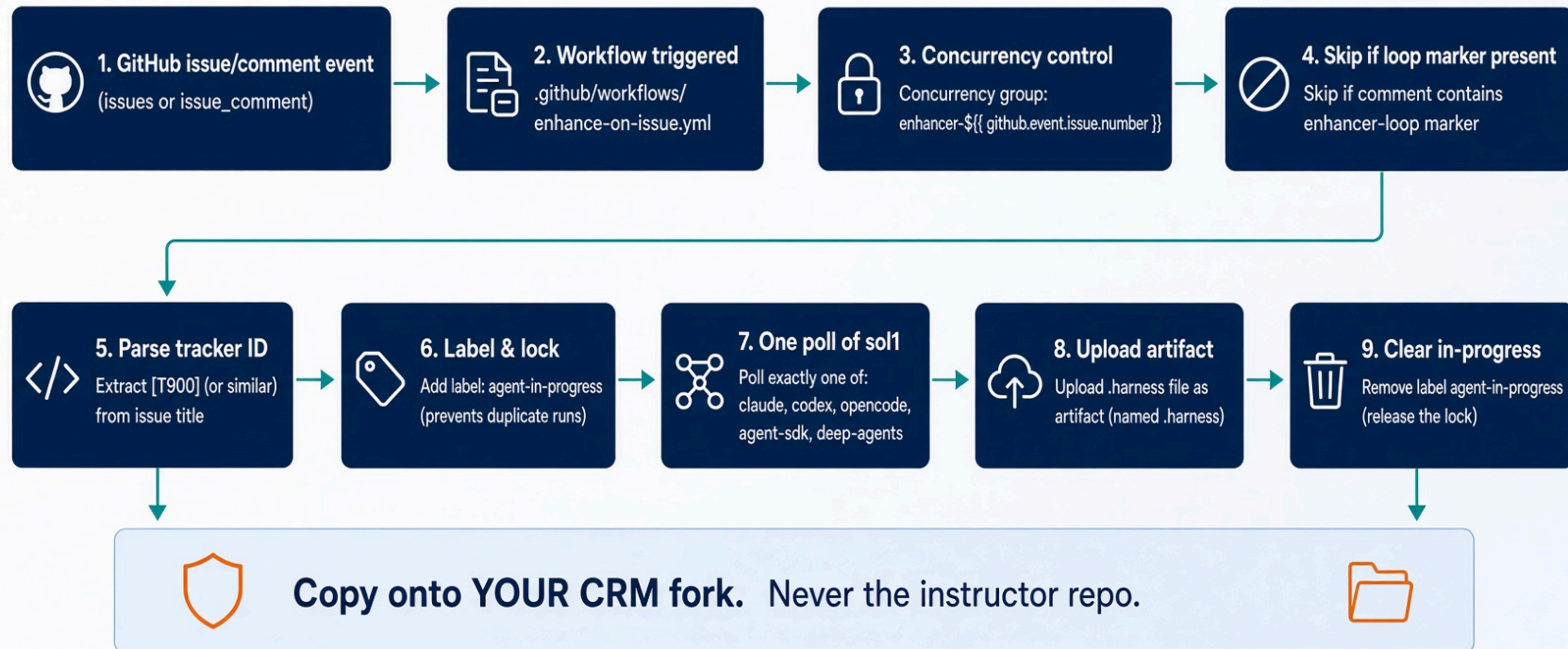
The trigger starts one poll. The enhancer exits. The workflow does not grade the ticket, merge, or deploy.

Learning objectives

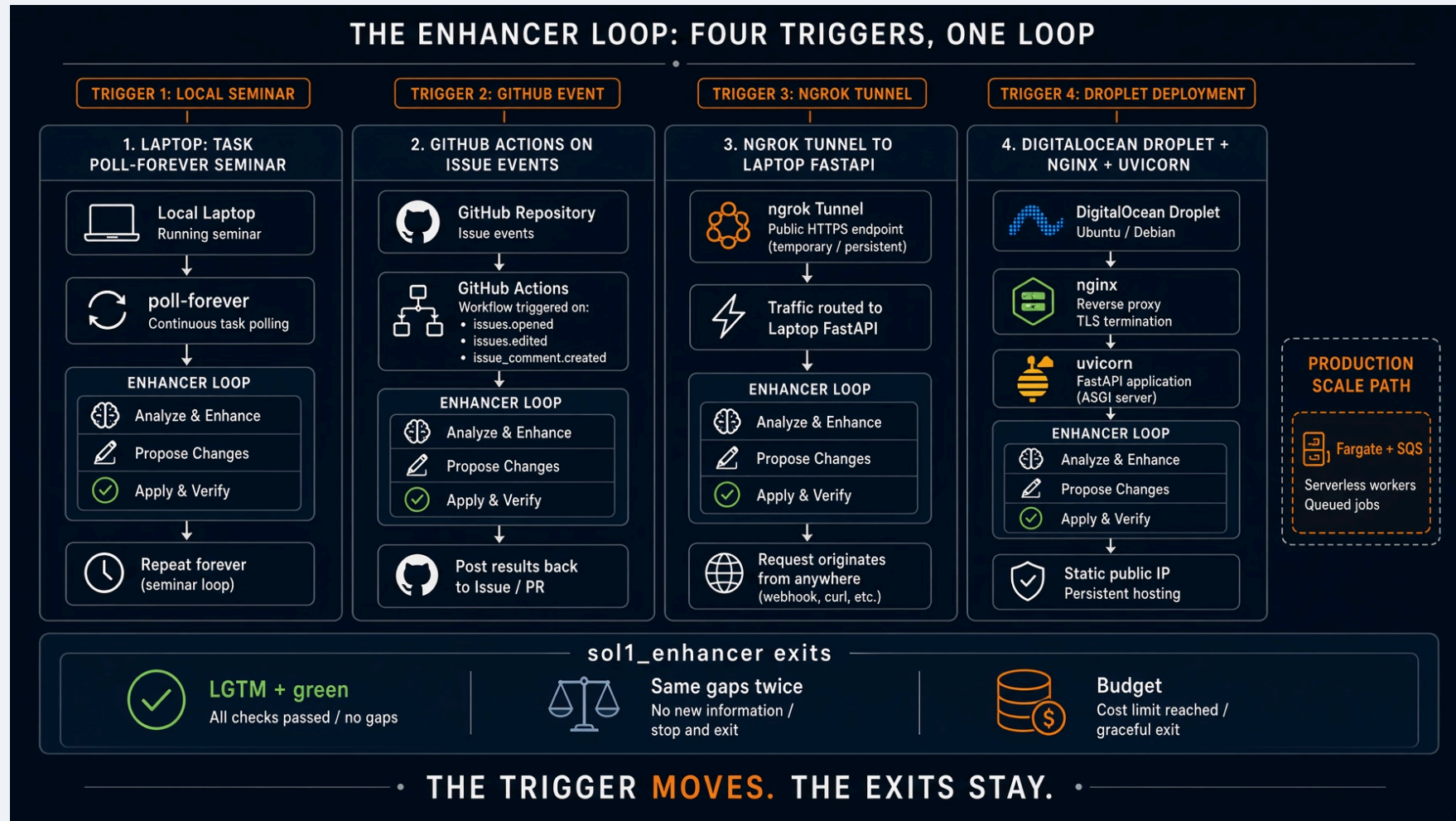
- Copy the YAML onto the CRM fork
- Parse `[T900]` from the issue title
- Skip `<!-- enhancer-loop -->`
- Serialize with `concurrency.group` per issue
- Swap `ENHANCER_BACKEND` without rewriting jobs
- Know which backends do **not** belong on hosted runners

Starting architecture

GitHub Actions as the receiver



Four triggers. Same exits.



Actions is column two. The loop is still `sol1_*`.

Copy onto YOUR fork

```
mkdir -p .github/workflows  
cp path/to/reliable-agentlab/labs/lab1_enhancer/workflows/enhance-on-issue.yml \  
  .github/workflows/enhance-on-issue.yml
```

Do not enable this on the shared instructor repo. It comments on issues and can spend model tokens.

Permissions. Least you need.

```
permissions:  
  issues: write  
  contents: read  
  actions: read
```

Grant `contents: write` only if a later job will push. The enhancer edits issue bodies and labels, not app code.

A workflow that merges has left the lab.

Concurrency per issue

```
concurrency:  
  group: enhancer-${github.event.issue.number || github.event.inputs.ticket}  
  cancel-in-progress: false
```

Two events on the same ticket must not chew it twice. Do not cancel in progress. A mid-poll kill leaves labels and a half-written candidate.

Skip the loop's own comments

```
if: >
  github.event_name == 'workflow_dispatch' ||
  github.event_name == 'issues' ||
  (github.event_name == 'issue_comment' &&
    !contains(github.event.comment.body, '<!-- enhancer-loop -->'))
```

Filter on the marker, never the author. The job runs as `github-actions[bot]`. A human LGTM must still count.

Ticket id from the title

```
id="$(printf '%s\n' "$TITLE" | grep -oE '\[T[0-9]+\]' | head -1 | tr -d '[]')"
if [ -z "$id" ]; then
    id="T${NUMBER}"
fi
```

Seed titles look like `[T900] Search crashes on an empty query`. If the title has no id, `T{issue_number}` matches `HOW_TO_RUN` for issues opened in the UI.

Backend matrix

ENHANCER_BACKEND	FOLDER	HOW THE JOB STARTS IT
claude (default)	sol1_enhancer	task run -- --ticket "\$TICKET"
codex	sol1_enhancer_codex	same
opencode	sol1_enhancer_opencode	same
agent-sdk	sol1_enhancer_agent_sdk	python3 loop.py --once ...
deep-agents	sol1_enhancer_deep_agents	python3 loop.py --once ...

Set repository variable ENHANCER_BACKEND. Secret ANTHROPIC_API_KEY for SDK and Deep Agents.

What does not belong on hosted runners

Grok Build. Trust prompt. Local plugin shims. `task trust` is a laptop step.

Keep Grok on a Droplet (`ext_5`) or a laptop (`ext_2`).

Codex needs `codex` on the runner. If the image does not have it, use Agent SDK or Claude Code.

In-progress and the second belt

```
gh issue edit "$NUMBER" --add-label agent-in-progress  
# ... one poll ...  
gh issue edit "$NUMBER" --remove-label agent-in-progress # if: always()
```

Stop at `AGENT_MAX_ATTEMPTS` (default 3) by counting `agent-attempts-*` labels. The loop already has a round budget. This is a second belt, at the trigger.

Two checkouts

```
- uses: actions/checkout@v4          # the CRM fork, $GITHUB_WORKSPACE
- uses: actions/checkout@v4
  with:
    repository: ${vars.ENHANCER_REPO || github.repository}
    path: .enhancer
```

`--repo "$GITHUB_WORKSPACE"` points the enhancer at the CRM files. The enhancer code lives in `.enhancer`.

Run one poll

```
case "$ENHANCER_BACKEND" in
  claude)    cd solutions/sol1_enhancer && task run -- --ticket "$TICKET" --repo "$TARGET" ;;
  agent-sdk) cd solutions/sol1_enhancer_agent_sdk && python3 loop.py --once --repo "$TARGET" --ticket "$TICKET" ;;
  deep-agents) cd solutions/sol1_enhancer_deep_agents && python3 loop.py --once --repo "$TARGET" --ticket "$TICKET" ;;
esac
```

Upload `.harness/last-enhancer-*.json` as an artifact. That is how you read the last score at 2 a.m.

What does not change

Three exits, no fourth: LGTM plus a green rubric, same gaps twice, budget spent.

Labels enhanced, ready, needs-human. Judge still has no write tool. `check_fields.py` still computes ready.

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Verify

1. Push the workflow to **your** fork
2. Open [T900] Search crashes on an empty query
3. Actions tab shows enhance. Issue gets enhanced or a marked comment
4. Comment LGTM as yourself. Next event sets ready only if the rubric is already green
5. A second event with no new human comment must be a no-op. If it posts again, the marker filter is missing

Troubleshooting

SYMPTOM	CAUSE	FIX
Never fires	workflow on the instructor repo	copy YAML to your fork
Loops	marker filter missing	skip <code><!-- enhancer-loop --></code>
404 on issues	token cannot write issues	issues: <code>write</code> , or a PAT for a private CRM
Grok skill not found	hosted runner	use <code>claude / agent-sdk / deep-agents</code>
No artifact	harness path wrong	upload <code>**/.harness/last-enhancer-*.json</code>

When Actions is the wrong tool

Laptop demo: extra credit 2, ngrok.

Permanent cheap box: extra credit 5, Droplet.

Long-running model calls, private VPC, or you already live on AWS: Fargate. Next deck.

If you stall, return to `task run --from labs/lab1_enhancer`. That is the Saturday path.

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Recap

Takeaways

1. Copy onto the CRM fork.
2. Trigger starts. Loop stops.
3. Marker, not author.
4. Concurrency per issue. Never cancel mid-poll.
5. Grok stays off hosted runners.

Closing line. The workflow is a doorbell. It is not the loop.

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